Zarif Hossain

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Education

Georgia Institute of Technology — Atlanta, GA
M.S. in Computer Science — *Jan 2026 – May 2027*

University of California, Irvine — Irvine, CA
B.S. Computer Science & Engineering — *Sept 2021 – Jun 2025*

- **Coursework:** Operating Systems, Data Structures & Algorithms, Networks, AI, Embedded Software, Compilers

Experience

Advanced Bionics — Santa Clarita, CA
Software Engineering Intern — *Jun 2024 – Mar 2025*

- Developed cross-platform medical app (Swift/Kotlin) with real-time audio DSP and REST synchronization for cochlear diagnostics, serving 1,000+ users across iOS and Android.
- Automated CI/CD via TeamCity + Fastlane and optimized audiometric algorithms using SciPy, cutting release cycles by 50% and test evaluation time by 25%.

Edwards Lifesciences — BEAT Lab — Irvine, CA
Software Engineering Undergraduate Researcher — *Jun 2023 – Jun 2025*

- Built an interactive Voilà interface for the MONARCH cardiac simulation engine enabling real-time hemodynamic tuning, visualization, and intuitive interpretation of model responses.
- Developed pandas/NumPy ETL pipelines reducing clinical dataset preprocessing time by 85%.
- Contributed to MONARCH Python package paper submitted to Journal of Open Source Software; developed validation workflows, numerical model checks, and comprehensive developer documentation.

UC Irvine CubeSat — Irvine, CA
Avionics Software Engineer — *Sept 2024 – Jun 2025*

- Developed FreeRTOS flight software with deterministic scheduling, IPC queues, and subsystem state management coordinating 12+ real-time avionics tasks.
- Implemented DMA-based I2C/SPI drivers with interrupt-driven IMU/magnetometer sampling for stable telemetry.

Projects

Fantasy Sports Revolution | MERN, Redis, Python, XGBoost

- Built full-stack fantasy football SaaS platform serving 100+ users with Auth0 authentication, Stripe payment processing, and real-time lineup optimization.
- Trained XGBoost regression model on 500K+ NFL player samples to generate 10K+ weekly statistical projections.
- Optimized API response times by 600ms through Redis caching layer and MongoDB query tuning; architected denormalized roster schema enabling sub-100ms lineup queries at scale.

OrbitOps | Rust, Kafka, ClickHouse, PyTorch, GraphQL

- Built Rust telemetry pipeline processing 10K+ events/min with zero-copy parsing and adaptive backpressure
- Deployed PyTorch LSTM autoencoder as gRPC microservice achieving 94% F1 score for anomaly detection.
- Implemented ClickHouse GraphQL API reducing query latency from 2.5s to 180ms; containerized multi-service stack with Docker + Kubernetes for scalable deployments.

RaceCraft AI | Spring Boot, Java, Python, AWS

- Built motorsports strategy simulation engine processing F1 telemetry datasets; modeled tire degradation curves and fuel consumption using polynomial regression on 100+ race samples.
- Implemented dynamic programming + Monte Carlo simulation for optimal pit-stop timing, improving strategy accuracy by 28% across 5,000+ simulated races.
- Deployed Spring Boot microservices on AWS EC2 with Redis caching, reducing strategy-query latency by 70%.

Technical Skills

Languages: Python, C/C++, Rust, Java, JavaScript/TypeScript, Swift, Kotlin, SQL
Frameworks: React, Node.js, Express, Spring Boot, FastAPI, GraphQL, PyTorch, scikit-learn
Tools/Systems: Docker, Kubernetes, AWS (EC2, S3), Kafka, FreeRTOS, Git, GitHub, TeamCity, TCP/UDP
Embedded/Hardware: Arduino, Verilog, VHDL, I2C/SPI, TCP/UDP, STM32
Databases: PostgreSQL, MongoDB, Redis, ClickHouse